



Govt. Graduate College, Pasrur
University of the Punjab, Lahore

Final Year Project Report
Department Management System
(The DMS Website)

Department Management System (The DMS Website)



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Project Advisor: Mam Ayesha

Submitted by:

Name of Student

Roll Number

Zainab Bibi

093554

Ghulam Khadija

093615

Bisma Shehzadi

093635

Statement of Submission

This is to certify that **Zainab Bibi (093554), Ghulam Khadija (093615) and Bisma Shehzadi (093635)**, under the supervision of **Prof.Ayesha**, has successfully completed the documentation of this Final Year Project at Govt.Graduate College, Pasrur, as a partial fulfillment of the requirement for the degree of BS in computer science.

Name of Student	Roll Number
Zainab Bibi	093554
Ghulam Khadija	093615
Bisma Shehzadi	093635

Project Supervisor

Mam Ayesha

Lecturer (V)

Govt. Graduate College Pasrur, Sialkot

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Table of Contents

1. INTRODUCTION	9
1.1. Background.....	9
1.2. Problem Statement:	9
1.3. Stakeholders & Interests	9
1.4. Objectives.....	10
1.6. Nexi Overview (AI Chatbot)	11
2. LITERATURE REVIEW	13
2.4. Technologies, Tools, and Libraries	14
2.5. Work Division	14
3. REQUIREMENTS ANALYSIS.....	16
3.2. Requirement Identifying Technique.....	17
3.3. Functional Requirements	18
3.3. Non-Functional Requirements	22
3.4. External Interface Requirements.....	23
4. USE CASE ANALYSIS.....	27
4.1. Use Case 1 (Customer Registration).....	27
4.2. Use Case 2 (Teacher Registration)	27
4.3. Use Case 3 (User Login – Student & Teacher)	28
4.4. Use Case 4 (Mark Attendance)	28
4.5. Use Case 5 (View Attendance)	29
4.6. Use Case 6 (Upload Grades)	29
4.7. Use Case 7 (View Grades).....	30
4.8. Use Case 8 (Ask Query via NEXI Chatbot)	30
4.9. Use Case 9 (Admin Manages Users).....	31
4.10. Use Case 10 (Upload Documents)	31
4.11. Use Case 11 (Download Documents)	32
4.12. Use Case 12 (Submit Leave Request)	32
4.13. Use Case 13 (Respond to Leave Request)	33
4.14. Use Case 14 (Notifications Management).....	33
4.2. Student Use Case Diagram.....	34
4.3. Teacher Use Case Diagram.....	34

4.4. Admin Use Case Diagram	35
5. SYSTEM DESIGN.....	37
5.1. Design Considerations.....	37
5.2. Requirements Traceability Matrix	38
5.3. Architectural Design	39
5.3.2. Database Diagram.....	39
5.3.3. ER Diagram	40
Design Models	41
5.4.1. Class Diagram	41
5.4.3. State Diagrams:.....	45
5.4.3. User Interface Design	48
6. IMPLEMENTATION	51
6.1. Development Tools and Environment	51
6.2. Modules Implemented	52
7. TESTING	63
8. SYSTEM CONVERSION.....	68
8.1. Conversion Method.....	68
8.2. Training: User Guide for Core Modules	68
8.2.1. Student Panel.....	68
8.2.2. Faculty Panel.....	68
8.2.3. Admin Panel	68
8.3. Challenges and Solutions.....	69
9. CONCLUSION.....	71
9.1. Evaluation.....	71
9.2. Traceability Matrix	71
Summary.....	72
References	72

Table of Figures

Figure 4.1: Student Use Case Diagram	34
Figure 4.2: Teacher Use Case Diagram	34
Figure 4.3: Admin Use Case Diagram	35
Figure 5.1: Database Diagram	39
Figure 5.2: ER Diagram	40
Figure 5.3: Class Diagram	41
Figure 5.4: Student Sequence Diagram	42
Figure 5.5: Teacher Sequence Diagram	43
Figure 5.6: Admin Sequence Diagram	44
Figure 5.7: Student State Diagram	45
Figure 5.8: Teacher State Diagram	46
Figure 5.9: Admin State Diagram	47

Chapter 1

Introduction

1. INTRODUCTION

1.1 Background

Many educational institutions still manage student records, attendance, and grades manually or semi-digitally, causing delays, errors, and duplicated data. The Department Management System provides a **centralized, secure, web-based platform**, including **leave applications**, role-based notifications, and admin approvals to improve communication and efficiency.

1.2 Problem Statement:

Manual or semi-digital systems increase workload, delay communication, and make tracking records difficult. A centralized system is required to provide **real-time updates, notifications, secure approvals**, and efficient academic record management.

1.3 Stakeholders & Interests

1.3.1 Students

- View academic records, timetables, and grades
- Download academic documents
- Submit **leave applications**
- Ask queries via AI chatbot NEXI
- Receive notifications about tasks, queries, or leave approvals

1.3.2 Teachers

- Manage attendance, grades, and student records
- Update timetables
- Apoint HOD
- Respond to student queries and leave requests
- Receive notifications for relevant queries and task updates

1.3.3 Admin (Department Administration)

- Approve teacher and student accounts (including HOD).
- Manage departments, subjects, and courses.
- Monitor notifications, task updates, and AI chatbot (NEXI) interactions.
- Admin login uses **pre-assigned credentials** with verification.

1.3.4 Institution / Department Management

- Monitor academic performance
- Review reports
- Ensure smooth departmental operations

1.3.5 System Administrator

- Maintain system security and backups.
- Provide technical support.

1.4 Objectives

1.4.1 Primary

- Build a role-based web platform for Students, Teachers, HOD, and Admin.
- Enable academic record management, timetable viewing, grade updates, document downloads, and **leave application submission**.
- Secure login for admin using pre-assigned credentials and verification.

1.4.2 Secondary

- Implement AI Chatbot (NEXI) for instant query resolution.
- Teachers respond to leave applications, queries, and task updates.
- Notifications sent to relevant users based on role.
- Develop a responsive, user-friendly interface.
- Ensure secure data storage and retrieval.

1.5 Scope

1.5.1 Functional Scope

1.5.1.1 Core Features

- Academic Records: Students view attendance, grades, and personal info.
- Leave Applications: Students submit requests; teachers/HOD respond.
- Notifications: Tasks, queries, and leave updates notify relevant users.
- AI Chatbot (NEXI): Provides instant query assistance.
- Role-based dashboards: Admin, Teacher, HOD, Students.

1.5.1.2 User Interface

- Role-based dashboards with clean, intuitive navigation
- Fully responsive layout, optimized for both desktop and mobile devices
- Dynamic content displayed according to user roles (Admin, Teacher, Student)

1.5.1.3 User Management

- Secure role-based **Register / Login** system
- Session-based login for stateful access control
- Secure data handling with proper access control based on roles

1.5.2 Technical Scope

1.5.2.1 Development Stack

- **Frontend:** React
- **Backend:** Node.js with Express.js
- **Database:** MongoDB for storing user data, academic records, and documents
- **Authentication:** Firebase Authentication for secure login and role management
- **Real-Time Communication:** AI Chatbot (NEXI) integration for instant student query support
- **File Handling:** Teachers can upload academic documents; students can download files
- **Future Integrations:** Advanced reporting, analytics, and additional modules for enhanced functionality

1.6 Nexi Overview (AI Chatbot)

- Helps students get answers to questions about attendance, grades, and documents
- Provides instant responses without waiting for teachers or admin
- Guides students on using the system easily
- Works securely without exposing sensitive data

Chapter 2

Literature Review

2. LITERATURE REVIEW

2.1 Related Work

Many institutions still use manual or semi-digital systems for managing student records, attendance, grades, and departmental operations. Some universities have online systems, but these often lack **centralized dashboards, AI support, leave applications, role-based notifications, and HOD management**. Existing solutions are complex or costly, and they don't address **real-time student-teacher-admin communication** efficiently. Our solution provides a **web-based, centralized platform** for managing academic data, leave requests, notifications, and AI-assisted query handling, suitable for small to medium educational departments.

2.2 Gap Analysis

2.2.1 Manual Record Handling

Gap: Managing attendance, grades, and student records manually is time-consuming, error-prone, and difficult to maintain over time.

2.2.2 Limited Communication Channels

Gap: Traditional systems do not provide a proper platform for student-teacher communication, leading to delays in updates or responses.

2.2.3 No AI-Based Query Support

Gap: Students cannot get instant answers to queries, increasing dependency on teachers and admin for every small question.

2.2.4 Lack of Centralized System

Gap: Data is scattered, making report generation, historical tracking, and system monitoring difficult for admin.

2.3 Proposed Solution

- Centralized web platform for students, teachers, HOD, and admin.
- Students can view records, grades, timetables, download documents, and submit **leave applications**.
- Teachers update academic records, respond to leave requests, and queries.
- HOD is highlighted but performs regular teaching duties.
- Admin approves accounts, monitors notifications, and oversees the system.
- AI Chatbot (NEXI) supports instant query resolution.
- Notifications sent to relevant users for queries, tasks, and leave approvals.

2.4 Technologies, Tools, and Libraries

Tools & Technologies	
Tools	Rationals
Simple React	Frontend development
Visual Studio Code	Development environment
Git & GitHub	Version control
Figma	UI/UX design
MongoDB	Database for student, teacher, and department data
Firebase Authentication	Secure role-based login
Node.js & Express.js	Backend and API development

2.5 Work Division

Sr.	Roll No.	Names	Role & Work Division
01	093554	Zainab Bibi	- Frontend development using Simple React - Designing and implementing Student, Teacher, and Admin dashboards - Backend support and integration with database - Overall system implementation and testing
02	093615	Ghulam Khadija	- AI Chatbot (NEXI) research and support - Working on chatbot flow, responses, and improvements - Providing suggestions for better user interaction - Assisting in testing and documentation
03	093635	Bisma Shehzadi	- AI Chatbot (NEXI) assistance and enhancement ideas - Suggesting improvements for student query handling - Helping in report writing and documentation - Supporting system testing and validation

Chapter 3

Requirements Analysis

3. REQUIREMENTS ANALYSIS

The Software Requirements Specification (SRS) for the Department Management System (DMS) defines the functional and non-functional requirements of the system. DMS is a centralized, web-based platform that manages students, teachers, attendance, academic records, and queries using role-based access, while improving efficiency, accuracy, and communication.

3.1 User Classes and Characteristics

User Class	User Characteristics
Student	• Register and login securely • View attendance, grades, and academic records • Submit leave requests • Use NEXI chatbot for instant assistance • Receive notifications about queries, tasks, and leave status
Teacher	• Login with authorized credentials • Mark and manage student attendance • Upload and update grades and academic records • Respond to student queries and leave requests • Receive notifications for assigned tasks and updates
HOD	• A teacher with a highlighted departmental role • Performs all regular teacher responsibilities • Handles departmental queries and leave requests • Receives department-related notifications
Admin	• Use pre-assigned login credentials • Approve student, teacher, and HOD accounts • Manage departments, subjects, and academic content • Monitor system activities and notifications • Maintain system security, backups, and provide technical support • Control overall system operations

3.2 Requirement Identifying Technique

The Use Case Technique is used to identify the functional requirements of the Department Management System (DMS). This technique helps in clearly defining interactions between users and the system by describing how students, teachers, and administrators perform different tasks. It ensures that all system functionalities are user-focused, easy to understand, and aligned with real departmental operations. The use case approach also helps in identifying system boundaries, roles, and expected outcomes effectively.

Use Case Name	Associated Functional Requirements
User Authentication	FREQ-1: Students, Teachers, and Admin can log in/register securely using role-based credentials.
View Attendance	FREQ-2: Students can view their attendance; teachers can mark/update attendance.
Academic Record Management	FREQ-3: Teachers can upload grades, results, and assignments; students can view their records.
Submit Query	FREQ-4: Students can submit academic or general queries to teachers.
Respond to Queries	FREQ-5: Teachers can view and respond to student queries via dashboard.
AI Chatbot Assistance	FREQ-6: Students can interact with NEXI chatbot for instant query resolution.
Manage Students	FREQ-7: Admin can add, update, or remove student records.
Manage Teachers	FREQ-8: Admin can add, update, or remove teacher records and assign subjects/semesters.
Announcements	FREQ-9: Admin can post notices and announcements for students and teachers.
Reports	FREQ-10: Admin and teachers can generate attendance and performance reports.
Monitor Queries & Chat	FREQ-11: Admin can view student-teacher interactions and chatbot queries for oversight.

3.3 Functional Requirements

The core features and operation are explained in Functional Requirements. They specify what the system should do to meet user needs.

Field	Description
ID	FREQ-1
Title	User Authentication
Requirements	System shall allow Admin, Teacher, and Student to register and login using secure credentials.
Source	All Users
Priority	High

Field	Description
ID	FREQ-2
Title	Role-Based Access
Requirements	System shall provide separate dashboards based on user roles (Admin, Teacher, Student).
Source	System
Priority	High

Field	Description
ID	FREQ-3
Title	Student Management
Requirements	Admin shall be able to add, update, view, and delete student records.
Source	Admin
Priority	High

Field	Description
ID	FREQ-4
Title	Teacher Management
Requirements	Admin shall manage teacher profiles and assign subjects/semesters.
Source	Admin
Priority	High

Field	Description
ID	FREQ-5
Title	Attendance Management
Requirements	Teachers shall mark and update attendance; students can view their attendance.
Source	Teacher / Student
Priority	High

Field	Description
ID	FREQ-6
Title	Academic Record Management
Requirements	Teachers shall upload marks, results, and assignments; students can view records.
Source	Teacher / Student
Priority	High

Field	Description
ID	FREQ-7
Title	Query Submission
Requirements	Students shall submit academic or general queries through the system.
Source	Student
Priority	Medium

Field	Description
ID	FREQ-8
Title	Query Response
Requirements	Teachers shall respond to student queries via dashboard.
Source	Teacher
Priority	Medium

Field	Description
ID	FREQ-9
Title	AI Chatbot (NEXI)
Requirements	System shall provide AI-based chatbot support for instant student queries.
Source	Student
Priority	Medium

Field	Description
ID	FREQ-10
Title	Announcements Management
Requirements	Admin shall create and publish notices for students and teachers.
Source	Admin
Priority	Medium

Field	Description
ID	FREQ-11
Title	Report Generation
Requirements	System shall generate attendance and academic performance reports.
Source	Admin / Teacher
Priority	Medium

Field	Description
ID	FREQ-12
Title	Data Backup
Requirements	System shall maintain secure backup of academic and user data.
Source	System
Priority	High

3.3 Non-Functional Requirements

The quality attributes and performance constraints of the system are explained in Non-Functional Requirements. They specify how the system should perform rather than what it does.

3.3.1 Reliability

ID	Description
NF-1	The system shall ensure high availability (99.9% uptime) using reliable backend services and database storage.
NF-2	The system shall maintain consistent performance with minimal failures during regular academic operations.

3.3.2 Usability

ID	Description
NF-3	The system interface shall be clean, simple, and user-friendly for Students, Teachers, and Admin dashboards.
NF-4	Students must be able to view attendance, grades, and timetable within a few easy navigation steps.

3.3.3 Performance

ID	Description
NF-5	The system shall support multiple concurrent users without noticeable performance degradation.
NF-6	Data retrieval and page loading time shall not exceed 2 seconds under normal conditions.

3.3.4 Security

ID	Description
NF-7	All users (Student, Teacher, Admin) must authenticate before accessing system features.
NF-8	Role-based access control shall be enforced to prevent unauthorized access to sensitive data.

3.4 External Interface Requirements

This section describes how the Department Management System (DMS) interacts with external software, hardware, and communication interfaces.

3.4.1 User Interface Requirements

This section defines the logical characteristics of the user interface to ensure consistency and ease of use:

1. The system shall provide a clean, simple, and user-friendly interface.
2. The system shall support both desktop and mobile web browsers with responsive design.
3. Minimum supported screen resolution shall be 720p (1280×720 pixels).
4. The user interface shall provide clear navigation menus based on user roles.

3.4.1.1 Student Dashboard Navigation

- **Home** – View dashboard overview, announcements, and notifications
- **Profile** – View personal and academic information
- **Attendance** – View attendance records
- **Timetable** – View class timetable
- **Grades** – View semester-wise results
- **Leave Application** – Submit and track leave requests
- **Documents** – Download academic documents
- **NEXI Chatbot** – Ask queries using AI chatbot
- **Notifications** – View updates about queries, tasks, and leave status
- **Support** – Contact department/admin

3.4.1.2 Teacher Dashboard Navigation

- **Home** – View dashboard summary and notifications
- **Attendance Management** – Mark and update student attendance
- **Grades Management** – Upload and update student grades
- **Student Records** – View assigned student information

- **Leave Requests** – View and respond to student leave applications
- **Queries** – Respond to student queries
- **Announcements** – Post class or department-related updates
- **Notifications** – View task updates, queries, and leave requests
- **Support** – Contact admin

3.4.1.3 Admin Dashboard Navigation

- **Users Management** – Approve, add, update, suspend, or delete students and teachers
- **Department Data** – Manage departments, semesters, subjects, and HOD assignment
- **Reports** – Generate attendance, academic, and leave reports
- **Announcements** – Post notices for students and teachers
- **Notifications Monitoring** – Monitor system notifications, queries, and task updates
- **Chat Monitoring** – Monitor student-teacher and chatbot interactions
- **System Settings** – Manage system configurations, security, and backups

3.4.2 Software interfaces

The system integrates with various software components for authentication, data storage, messaging, and file uploads.

External System	Description
Authentication Service	Provides secure login for students, teachers, and admin
Database (MongoDB / SQL)	Stores student records, attendance, grades, and documents
AI Chatbot (NEXI)	Provides instant query support to students
Backend Server	Handles business logic and data processing
Frontend Framework	Displays interactive and responsive user interfaces

3.4.3 Hardware Interfaces

Component	Description
Smartphone / Tablet / Laptop	The system runs on any modern web browser
Server Hardware	Hosts the application and database
Internal Storage	Used for temporary session and cache data

3.4.4 Communication Interfaces

Component	Description
Internet (Wi-Fi / Mobile Data)	Required for accessing the web-based system
HTTP/HTTPS	Used for secure data communication between client and server
AI Chat Interface	Enables interaction between users and NEXI chatbot

Chapter 4

Use Case Analysis

4. USE CASE ANALYSIS

4.1 Use Case 1 (Customer Registration)

Use Case ID	UC-1
Title	Student Registration
Purpose	Registering a new student
Actor	Student
Precondition	Student has internet access and a web browser
Main Success Scenario	1. Student clicks “Register” 2. Enters personal and academic details 3. Submits form 4. Account is created
Alternate Solution	Email already exists – error message shown
Postconditions	Student account is created
Includes	None
Extends	None

4.2 Use Case 2 (Teacher Registration)

Use Case ID	UC-2
Title	Teacher Registration
Purpose	Registering a new teacher
Actor	Teacher
Precondition	Internet access available
Main Success Scenario	1. Teacher fills registration form 2. Submits details 3. Account created (pending admin approval)
Alternate Solution	Invalid data – error shown
Postconditions	Teacher account created
Includes	None
Extends	End

4.3 Use Case 3 (User Login – Student & Teacher)

Use Case ID	UC-3
Title	User Login
Purpose	Authenticate users
Actor	Student, Teacher
Precondition	Valid credentials
Main Success Scenario	1. User enters login details 2. System verifies credentials 3. User redirected to dashboard
Alternate Solution	Invalid credentials – error shown
Postconditions	User is logged in
Includes	None
Extends	UC-1, UC-2

4.4 Use Case 4 (Mark Attendance)

Use Case ID	UC-4
Title	Mark Attendance
Purpose	Record student attendance
Actor	Teacher
Precondition	Teacher is logged in
Main Success Scenario	1. Teacher selects class 2. Marks attendance 3. Saves record
Alternate Solution	Class not found – error shown
Postconditions	Attendance saved
Includes	None
Extends	UC-3

4.5 Use Case 5 (View Attendance)

Use Case ID	UC-5
Title	View Attendance
Purpose	Allow students to view attendance
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student opens attendance page. 2. System displays attendance record.
Alternate Solution	No data available – message shown
Postconditions	Attendance viewed
Includes	None
Extends	UC-3

4.6 Use Case 6 (Upload Grades)

Use Case ID	UC-6
Title	Upload Grades
Purpose	Upload student grades
Actor	Teacher
Precondition	Teacher logged in
Main Success Scenario	1. Teacher selects subject 2. Uploads grades 3. Saves data
Alternate Solution	Invalid data – error shown
Postconditions	Grades stored
Includes	None
Extends	UC-3

4.7 Use Case 7 (View Grades)

Use Case ID	UC-7
Title	View Grades
Purpose	Allow students to see grades
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student opens grades section 2. Grades displayed.
Alternate Solution	Grades not uploaded – message shown
Postconditions	Grades viewed
Includes	None
Extends	UC-6

4.8 Use Case 8 (Ask Query via NEXI Chatbot)

Use Case ID	UC-8
Title	AI Chatbot Query
Purpose	Provide instant answers
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student opens NEXI chatbot 2. Asks query 3. Chatbot responds.
Alternate Solution	Chatbot unavailable – retry later
Postconditions	Query answered
Includes	None
Extends	UC-3

4.9 Use Case 9 (Admin Manages Users)

Use Case ID	UC-9
Title	User Management
Purpose	Manage students and teachers
Actor	Admin
Precondition	Admin logged in
Main Success Scenario	1. Admin views users 2. Adds/updates/deletes users
Alternate Solution	Action failed – error shown
Postconditions	User data updated
Includes	None
Extends	UC-3

4.10 Use Case 10 (Upload Documents)

Use Case ID	UC-10
Title	Upload Documents
Purpose	Upload academic files
Actor	Teacher
Precondition	Teacher logged in
Main Success Scenario	1. Teacher uploads document 2. File saved to system
Alternate Solution	Upload failed – retry
Postconditions	Document available
Includes	None
Extends	UC-3

4.11 Use Case 11 (Download Documents)

Use Case ID	UC-11
Title	Download Documents
Purpose	Allow students to download files
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student selects document. 2. Downloads file.
Alternate Solution	File not found
Postconditions	Document downloaded
Includes	None
Extends	UC-10

4.12 Use Case 12 (Submit Leave Request)

Use Case ID	UC-12
Title	Submit Leave Request
Purpose	Allow students to submit leave applications
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student opens leave application section 2. Enters leave details and reason 3. Submits leave request
Alternate Solution	Incomplete details – error message shown
Postconditions	Leave request submitted and sent to teacher/HOD
Includes	None
Extends	UC-3

4.13 Use Case 13 (Respond to Leave Request)

Use Case ID	UC-11
Title	Download Documents
Purpose	Allow students to download files
Actor	Student
Precondition	Student logged in
Main Success Scenario	1. Student selects document. 2. Downloads file.
Alternate Solution	File not found
Postconditions	Document downloaded
Includes	None
Extends	UC-10

4.14 Use Case 14 (Notifications Management)

Use Case ID	UC-14
Title	View Notifications
Purpose	Notify users about queries, tasks, and leave updates
Actor	Student, Teacher, HOD, Admin
Precondition	User logged in
Main Success Scenario	1. System generates notification 2. User opens notifications section 3. Views notification details
Alternate Solution	No notifications available – message shown
Postconditions	Notification viewed
Includes	None
Extends	UC-3

Student Use Case Diagram

The Student Use Case Diagram represents actions such as viewing attendance, grades, downloading documents, and using the NEXI chatbot.

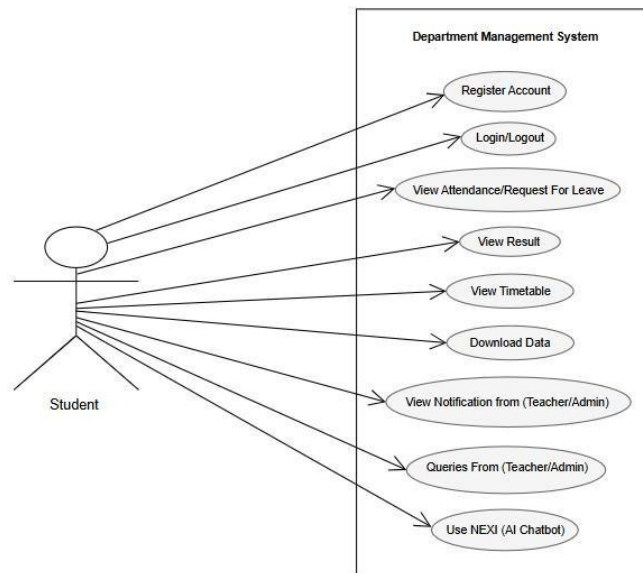


Figure 4.1: Student Use Case Diagram

Teacher Use Case Diagram

The Teacher Use Case Diagram shows tasks like marking attendance, uploading grades, documents, and responding to queries.

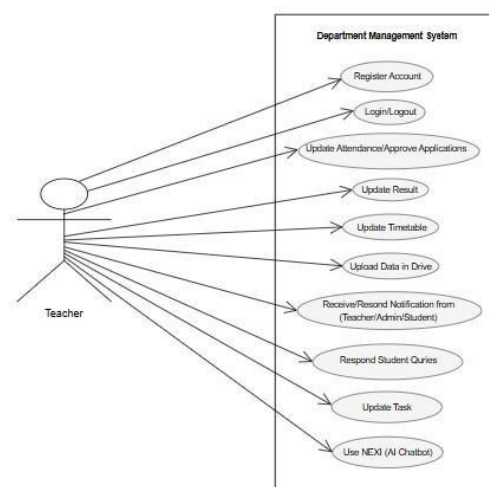


Figure 4.2: Teacher Use Case Diagram

Admin Use Case Diagram

The Admin Use Case Diagram defines responsibilities such as managing users, monitoring data, and maintaining system integrity.

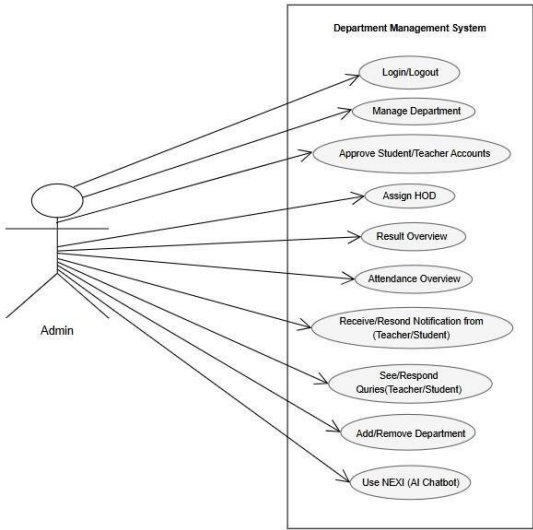


Figure 4.3: Admin Use Case Diagram

Chapter 5

System Design

5. SYSTEM DESIGN

The **Department Management System (DMS)** is designed as a centralized, web-based platform that enables efficient management of departmental activities involving **students, teachers, and administrators**. The system architecture follows a modular and scalable approach, with a user-friendly frontend and a secure backend connected to a centralized database.

The system supports **role-based access control**, where:

- **Students** can view attendance, academic records, announcements, and submit queries.
- **Teachers** can manage attendance, upload marks, share announcements, and respond to student queries.
- **Admin** oversees the entire system by managing users, departments, records, and monitoring system activities.

DMS aims to reduce manual workload, improve communication, and ensure accurate academic record management. The system is accessible via modern web browsers and provides a clean, responsive interface for better usability.

5.1 Design Considerations

5.1.1 Assumptions and Dependencies

1. Users have stable internet connectivity to access the system.
2. Students, teachers, and admin have basic computer literacy.
3. Academic data provided by teachers is accurate and timely.
4. The system is primarily designed for educational institutions operating within Pakistan.
5. Authentication services and database servers remain available during system operation.

5.1.2 Limitations

1. The system does not support offline access; all features require internet connectivity.
2. Advanced analytics and predictive reporting are not included in this phase.
3. AI chatbot responses are limited to predefined academic and departmental queries.
4. Integration with external learning platforms (LMS) is not supported currently.

5.1.3 Risks and Mitigation

Risks	Mitigation
Incorrect or fake user registrations	Role-based verification and admin approval
Data loss or server failure	Regular backups and cloud-based database
Misuse of communication channels	Chat monitoring and access control by admin
System overload during peak usage	Optimized queries and scalable backend

5.2 Requirements Traceability Matrix

The **Requirement Traceability Matrix (RTM)** for the **Department Management System (DMS)** establishes a clear relationship between identified functional requirements and their corresponding design components. This ensures that all system requirements are fully implemented and traceable throughout the system design.

Requirement ID	Requirement Description	Design Specification
FREQ-1	Students, teachers, and admin must be able to register and log in securely	Authentication Module
FREQ-2	Admin must be able to manage students and teachers	User Management Module
FREQ-3	Teachers must be able to mark and update attendance	Attendance Management Module
FREQ-4	Teachers must upload and manage student marks	Academic Records Module
FREQ-5	Students must be able to view attendance and results	Student Dashboard
FREQ-6	Students and teachers should communicate via chat or queries	Communication / Chat Module
FREQ-7	Admin must be able to post announcements and notices	Announcement Management
FREQ-8	Students can submit academic or departmental queries	Query Handling Module
FREQ-9	AI chatbot should answer common student queries	AI Chatbot Integration
FREQ-10	Admin can generate reports (attendance, performance)	Reporting Module
FREQ-11	Admin can monitor system activities and data	Admin Dashboard
FREQ-12	Students must be able to submit leave requests	Leave Management Module
FREQ-13	Teachers must be able to respond to student leave requests	Leave Response Module
FREQ-14	System must notify users about leave actions and system updates	Notification Module

5.3 Architectural Design

5.3.1 Data Design

The DMS uses **MongoDB**, a NoSQL database, to store and manage academic and administrative data efficiently. The DMS uses MongoDB, a NoSQL database, to store and manage academic and administrative data efficiently. The main collections include **Users, Students, Teachers, Attendance, Subjects, Results, Messages, Queries, LeaveRequests, and Notifications**. Each document is uniquely identified using ObjectIDs, and relationships are maintained through references (e.g., attendance linked to student and subject IDs, leave requests linked to student and teacher IDs). This flexible schema supports scalability, quick updates, real-time data access, and effective tracking of academic as well as leave-related activities.

Database Diagram

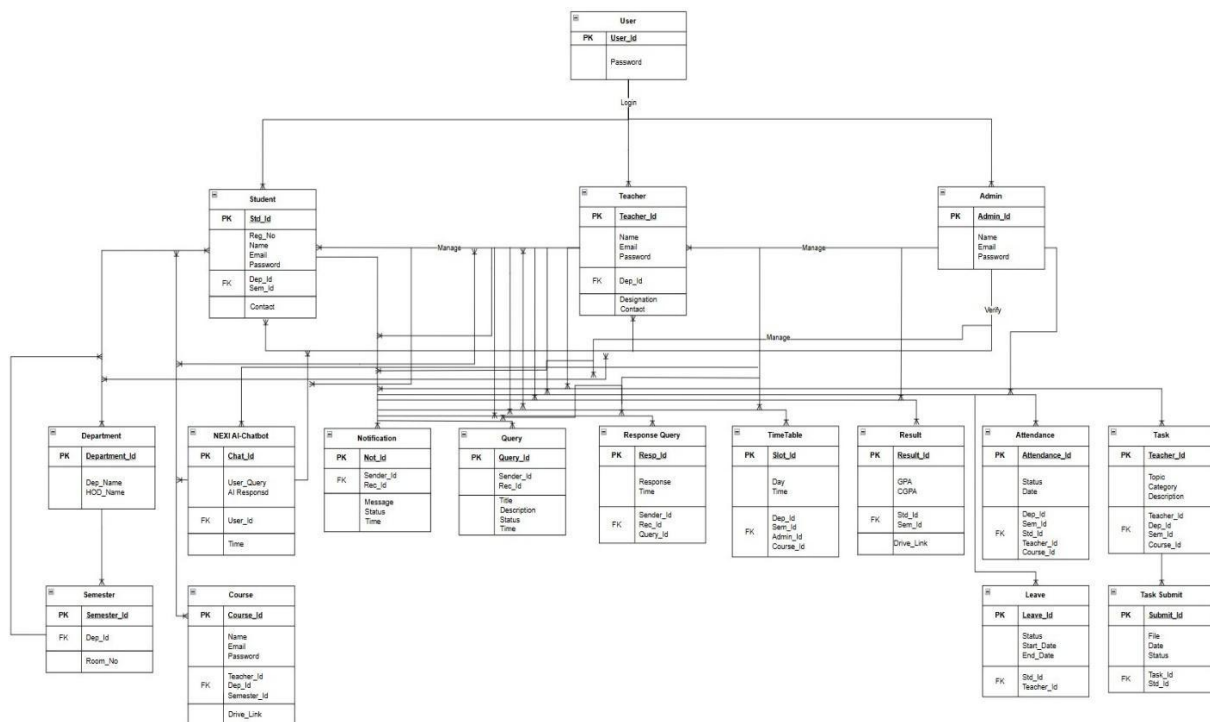


Figure 5.1: Database Diagram

ER Diagram

The Entity Relationship (ER) Diagram of the Department Management System (DMS) illustrates the logical structure of the database and the relationships between key entities such as **Admin**, **Department**, **Student**, **Teacher**, **Course**, **Semester**, **Attendance**, **Result**, **LeaveRequest**, and **Notification**. Each entity includes essential attributes to store academic, administrative, and leave-related information, while the relationships define interactions such as departments managing multiple students and teachers, teachers handling multiple courses, students submitting leave requests, and notifications being generated for relevant system events. The ER diagram helps in understanding the data model clearly and supports effective database design and backend implementation.

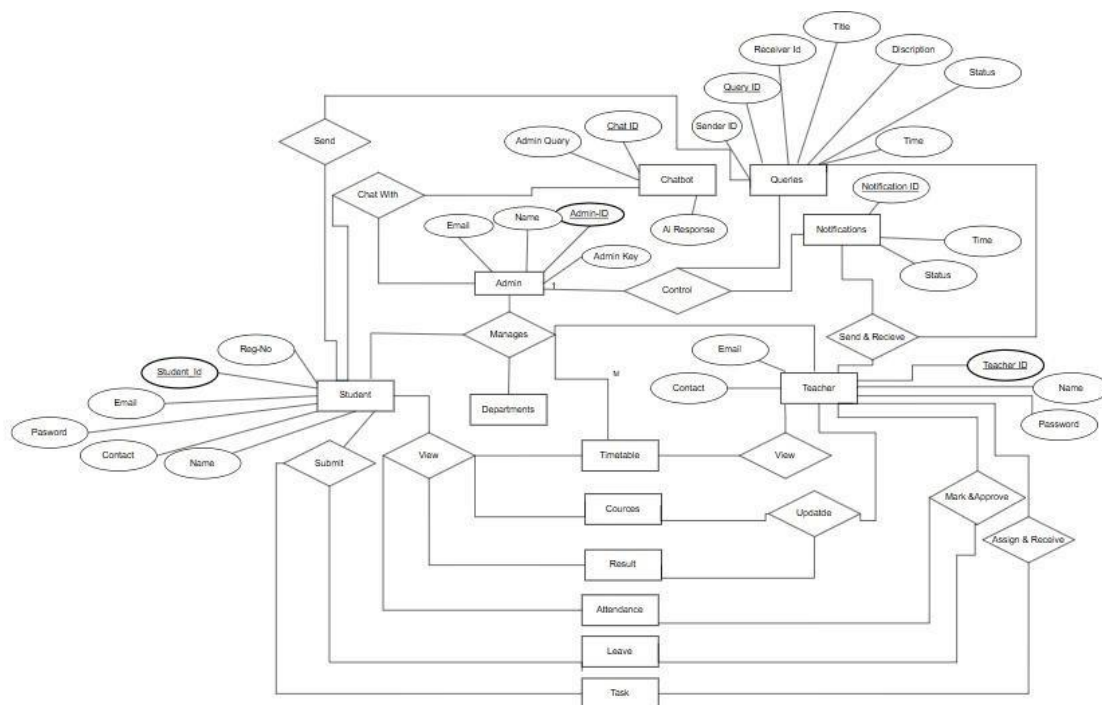


Figure 5.2: ER Diagram

Design Models

The system design of the Department Management System (DMS) follows an object-oriented development approach to ensure modularity, scalability, and ease of maintenance. To represent both the structural and behavioral aspects of the system, the following design models have been used:

1. Class Diagram

This diagram depicts the static structure of the system by illustrating core classes such as **Admin**, **Department**, **Student**, **Teacher**, **Course**, **Semester**, **Attendance**, **Result**, **LeaveRequest**, and **Notification**. It defines class attributes, methods, and relationships, providing a clear view of how academic, administrative, and leave management components are interconnected within the system.

2. Interaction Diagram (Sequence)

Sequence diagrams are used to model the dynamic interactions between system objects for key operations such as student registration, course assignment, attendance marking, result entry, **leave request submission**, **leave approval**, and administrative approvals. These diagrams help visualize the sequence of messages and the flow of control during different user actions.

3. State Transition Diagram

This diagram represents the different states of major entities like **Student**, **Course**, **Attendance**, **Result**, and **LeaveRequest** and shows how they transition from one state to another based on events such as enrollment, attendance submission, leave approval/rejection, result approval, or record updates. It also illustrates login/logout states and access control for admin, teachers, and students.

Class Diagram

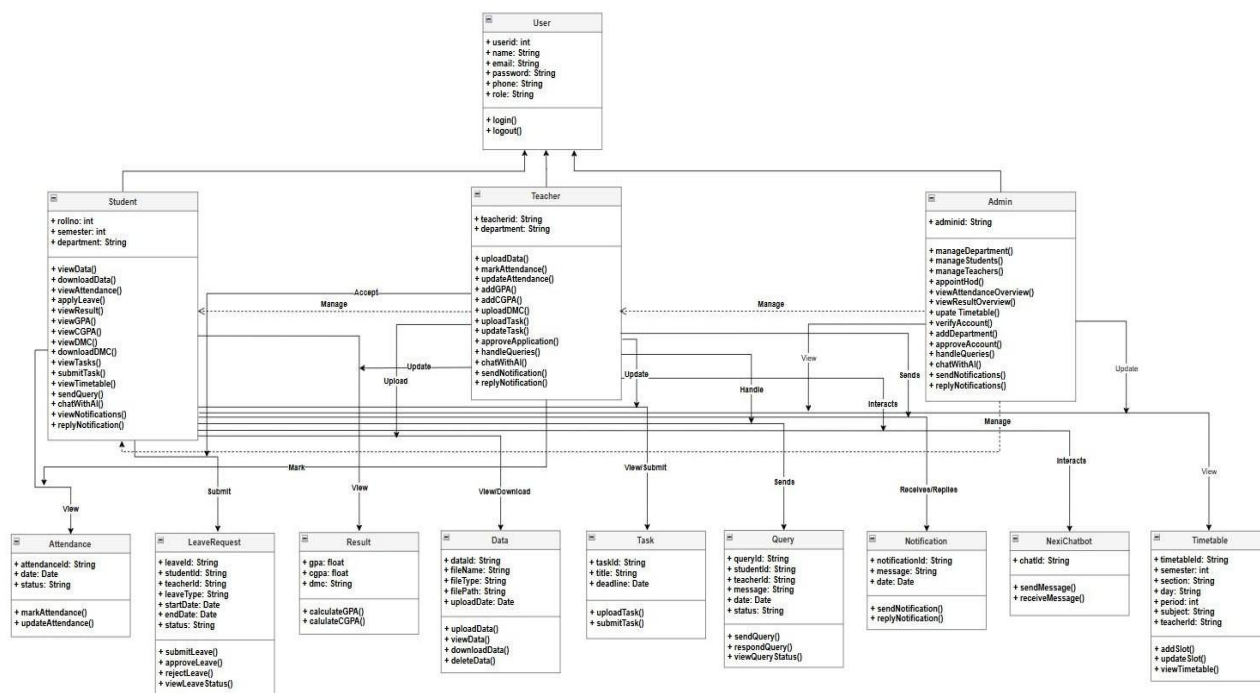


Figure 5.3: Class Diagram

5.4.1 Sequence Diagrams:

Student Sequence Diagram

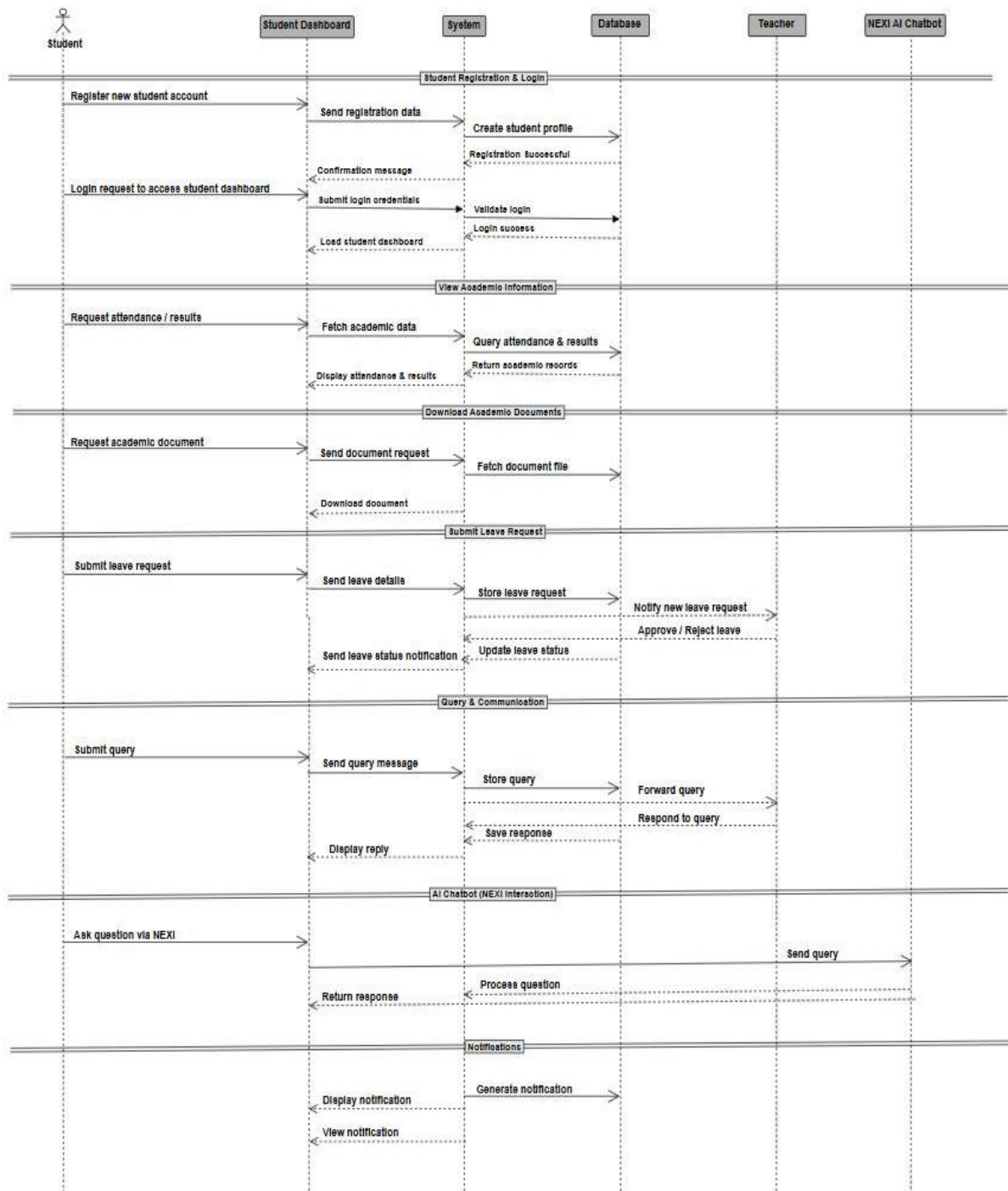


Figure 5.4: Student Sequence Diagram

Teacher Sequence Diagram

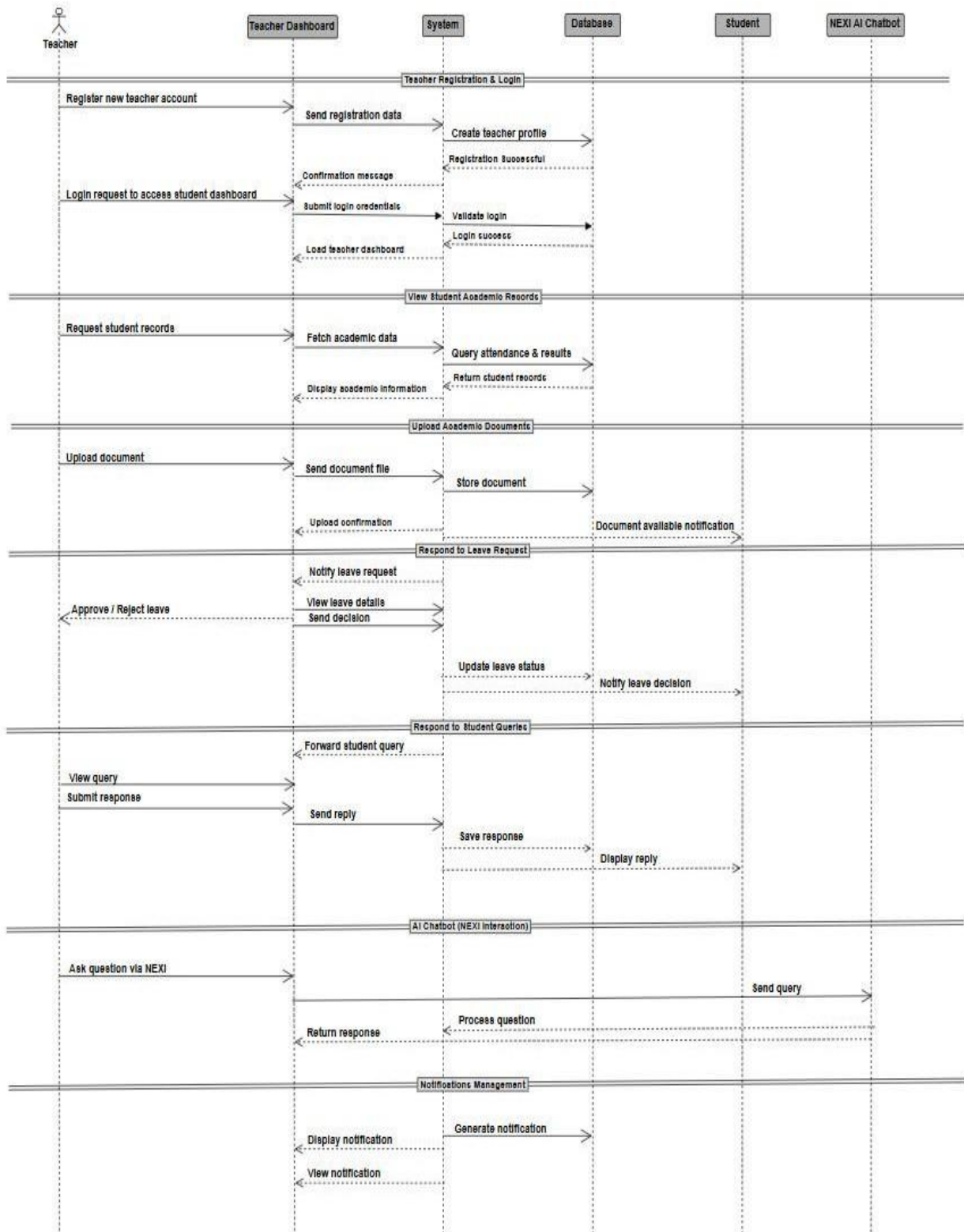


Figure 5.5: Teacher Sequence Diagram

Admin Sequence Diagram

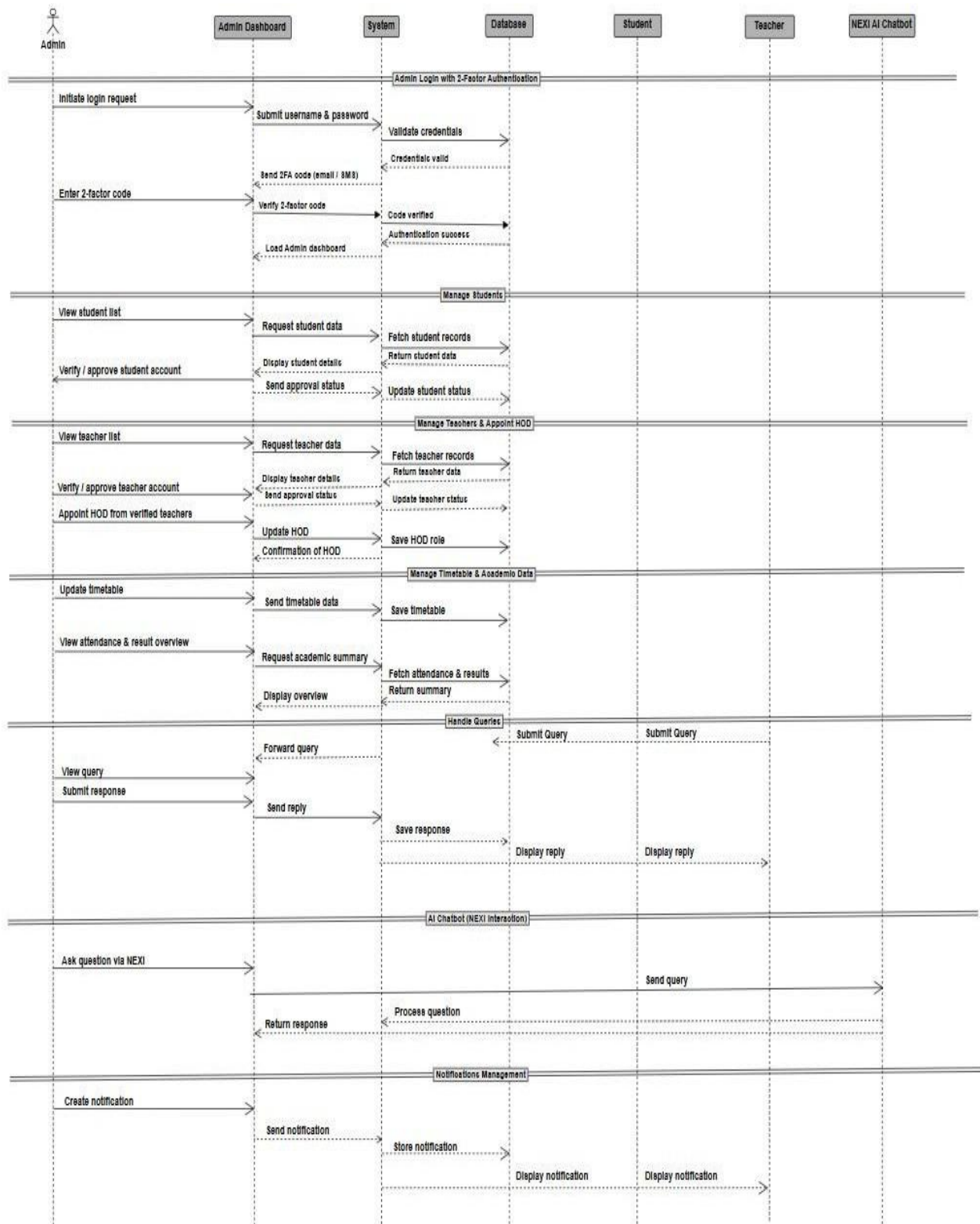


Figure 5.6: Admin Sequence Diagram

State Diagrams:

1. Student Panel

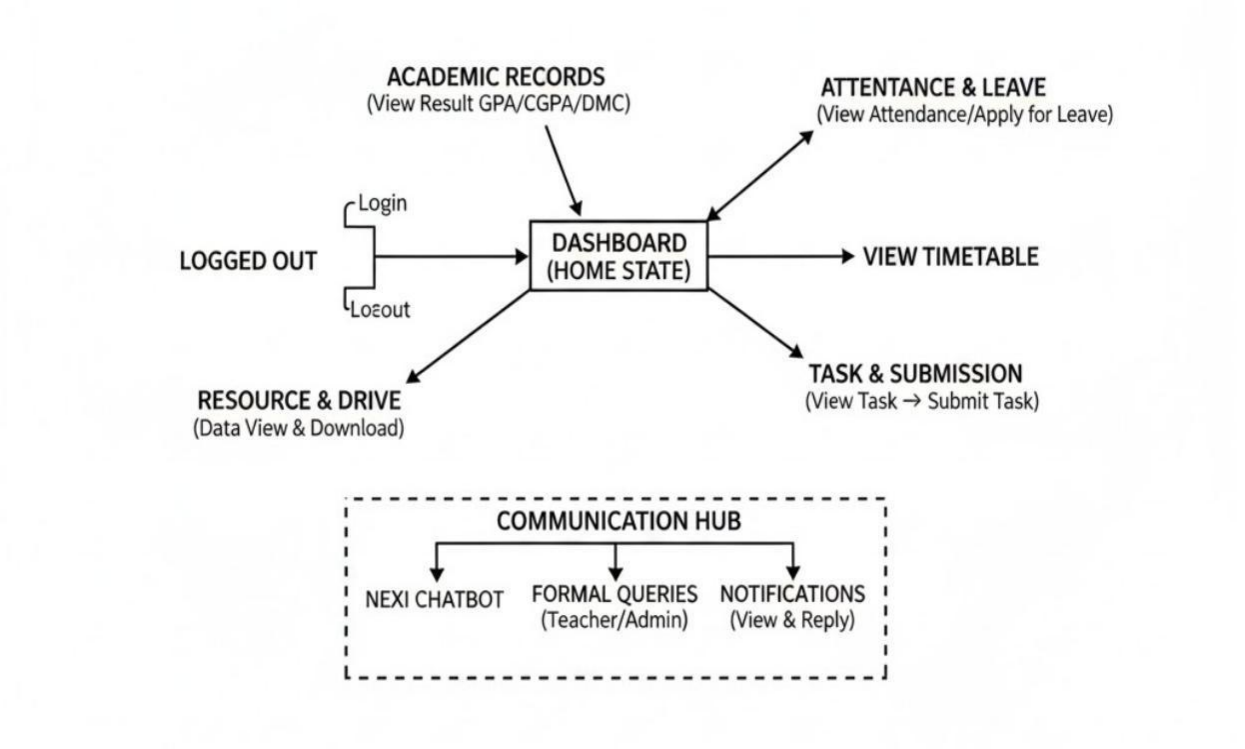


Figure 5.7: Student State Diagram

Teacher State Diagram

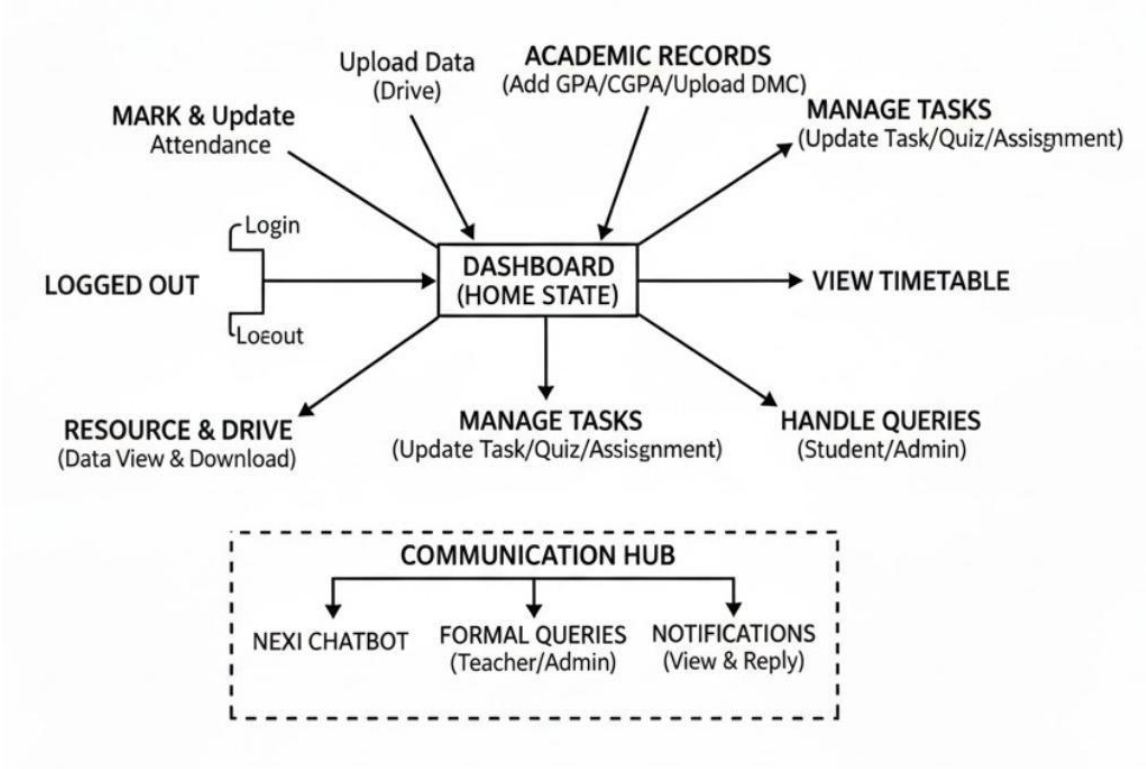


Figure 5.8: Student Sequence Diagram

Admin Dashboard

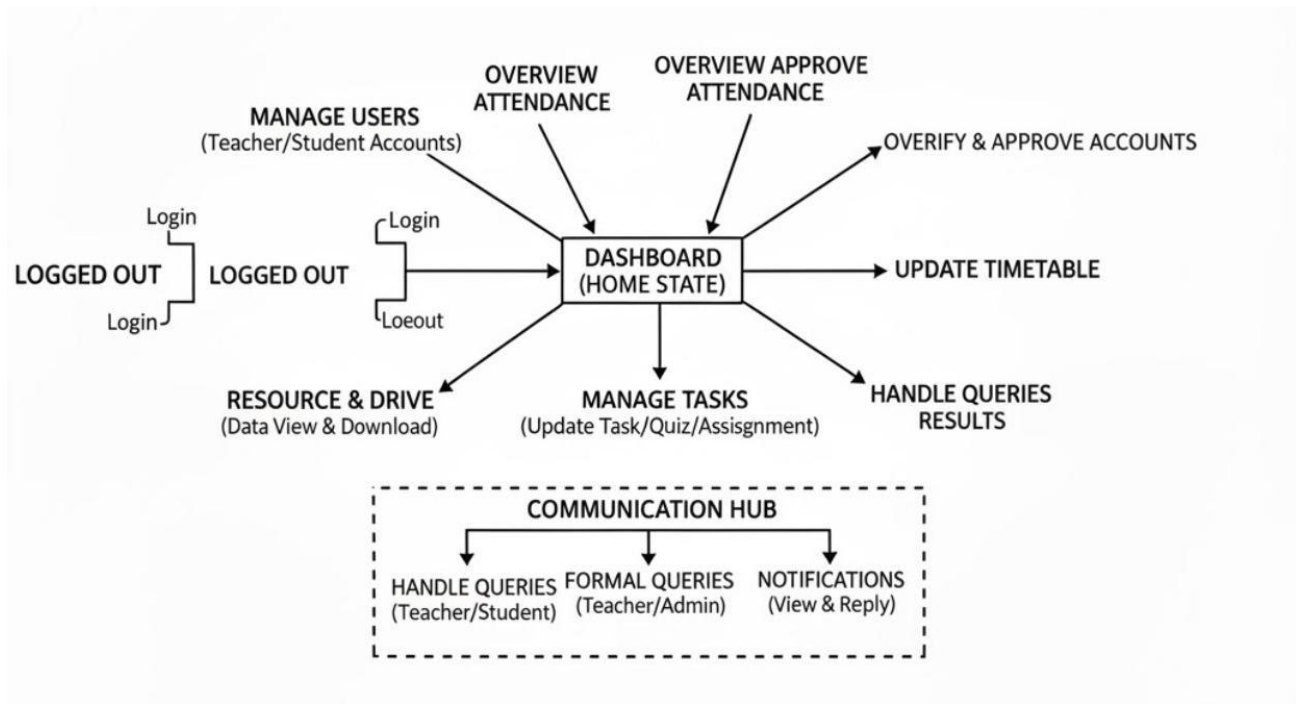


Figure 5.9: Admin Sequence Diagram

5.4.3 User Interface Design

The user interface of the Department Management System (DMS) is designed to be simple, responsive, and user-friendly to support academic and administrative tasks efficiently. Students can view their profiles, enrolled courses, attendance records, results, **submit leave requests**, and receive notifications through an intuitive dashboard. Teachers are provided with a dedicated interface to manage courses, mark attendance, upload results, **respond to leave requests**, and handle student queries. The admin panel offers full control over department management, user verification, course allocation, academic record monitoring, **leave oversight**, and notification management. The interface is designed using React and Tailwind CSS, ensuring consistent performance, accessibility, and responsiveness across different devices.

Here you see Figma design: [Click here](#)

5.5 Design Decisions

1. Approach Used

- **Object-Oriented Design (OOD)**

2. Reason

- **Modularity:** OOD divides the system into independent and reusable components such as Student, Teacher, Course, and Attendance, Leave Request .
- **Encapsulation:** Data and related operations are bundled within objects, improving system security and maintainability.
- **Real-World Mapping:** The system models real academic entities like departments, semesters, and users, making the design intuitive and easy to understand.

3. Technologies Used

1. **MongoDB** – For storing academic records, user data, attendance, and results.
2. **Node.js & Express.js** – For backend logic and API handling.
3. **React** – For building a dynamic and interactive frontend.
4. **JWT Authentication** – For secure user access and role-based authorization.

4. Reasons for Using These

1. Efficient Data Management with MongoDB

- Stores large volumes of academic and user data efficiently.
- Supports flexible schema for semester-wise and department-wise records.
- Scales easily with growing student and course data.

2. Secure and Scalable Backend

- Node.js and Express.js ensure fast request handling and smooth integration.
- JWT provides secure login and role-based access for admin, teacher, and student.

3. Seamless Frontend Integration

- React integrates smoothly with the backend APIs.
- Tailwind CSS enables responsive and consistent UI design.
- Ensures a smooth user experience across desktop and mobile devices.

Chapter 6 Implementation

6. Implementation

The Department Management System (DMS) was developed using **React.js** for the frontend and **Node.js** with **Express.js** for the backend. **MongoDB** was used as the database to store student, faculty, course, and attendance data. **Firestore Authentication** handles secure user login and registration for all roles. Students can view grades and attendance, while faculty can manage academic records and course materials. Admins handle system content, user roles, and department-wide reporting. The system includes a communication module for seamless interaction between faculty and students.

6.1. Development Tools and Environment

For the development of the Department Management System, various tools, frameworks, and technologies were selected to ensure efficient development and maintenance.

Tool	Version	Rationale
VS Code	v1.85.1	Primary IDE for coding and debugging.
GitHub	Cloud-Based	Facilitates version control and collaboration.
drawio	v24.x.x	Used for Gantt charts and project scheduling.
Figma	v6.0	Used for UI/UX design and prototyping.

Technologies	Version	Rationale
React.js	v18.2.0	Provides a modern and fast user experience.
Node.js	v20.10.0	Handles server-side logic efficiently.
Express.js	v4.18.2	Simplifies creation of RESTful APIs.
MongoDB	v6.0.13	Offers scalability for academic data storage.
Firebase	Cloud-Based	Handles secure authentication.

6.2. Modules Implemented

6.2.1. Authentication Module

- Registration and Login for Students and Faculty.
- Admin login with restricted access.
- Secure credential validation and role-based redirection.

6.2.2. Admin Panel Module

- Admin login to access the main dashboard.
- Manage all user accounts (Students & Faculty).
- Add, edit, or delete department announcements.
- Update static content (About, Policies, etc.).
- View and resolve reported issues or complaints.

6.2.3. Faculty Dashboard Module

- Access dashboard to manage course listings.
- Upload and manage course materials and syllabi.
- Input and update student grades.
- Take and manage student attendance.
- Communicate with students regarding academic queries.

6.2.4. Student Dashboard Module

- Register and log in to personal student portal.
- View grades and academic performance history.
- Check attendance status and course schedules.
- Access uploaded course materials.
- Contact faculty or support for academic issues.

6.2.5. Course & Attendance Management Module

- Allow faculty to manage course details.
- Real-time attendance tracking and status updates.
- Attendance reports and automated notifications.

6.2.6. Grade Management Module

- Faculty can record and edit marks for subjects.
- Students can view final and internal assessment results.

6.2.7. Communication Module

- Messaging system between faculty and students.
- Admin access to logs for moderation.

6.2.8. Review & Rating Module

- Students can provide feedback on course delivery.
- Admin can manage course-related reviews.

6.2.9. Issue Management & Support Module

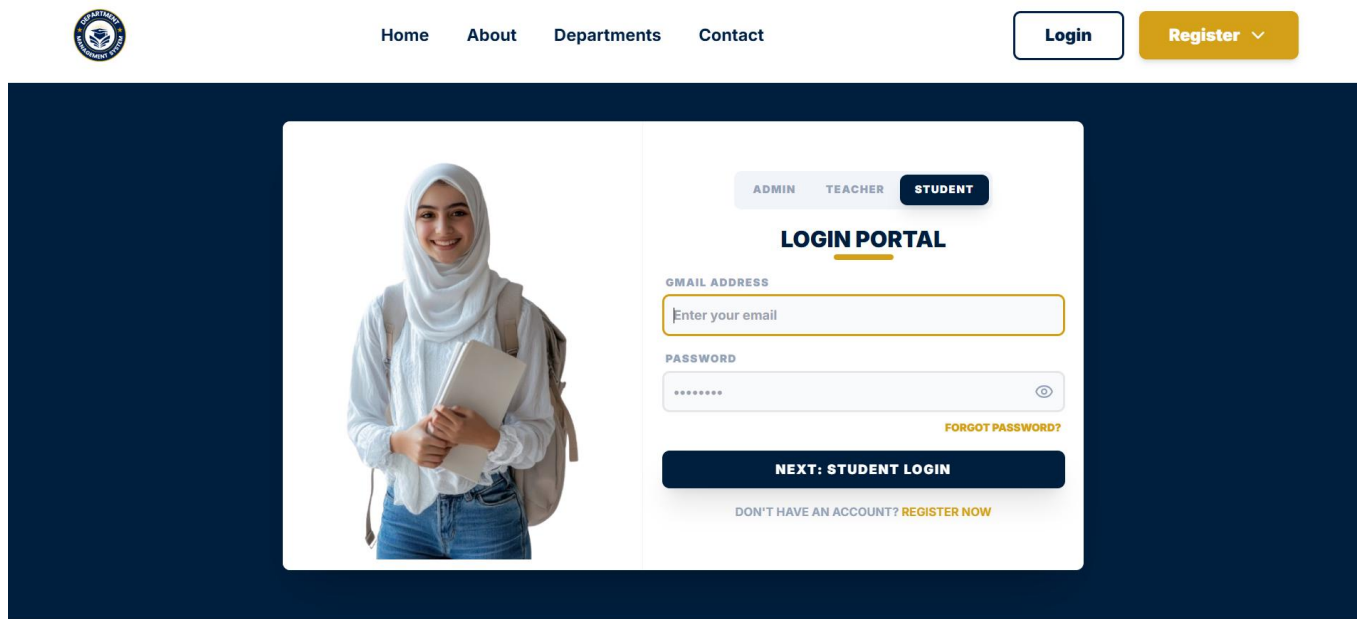
- Admin handles reported academic complaints.
- Ticket-based system for query resolution.

6.2.10. Database Management Module

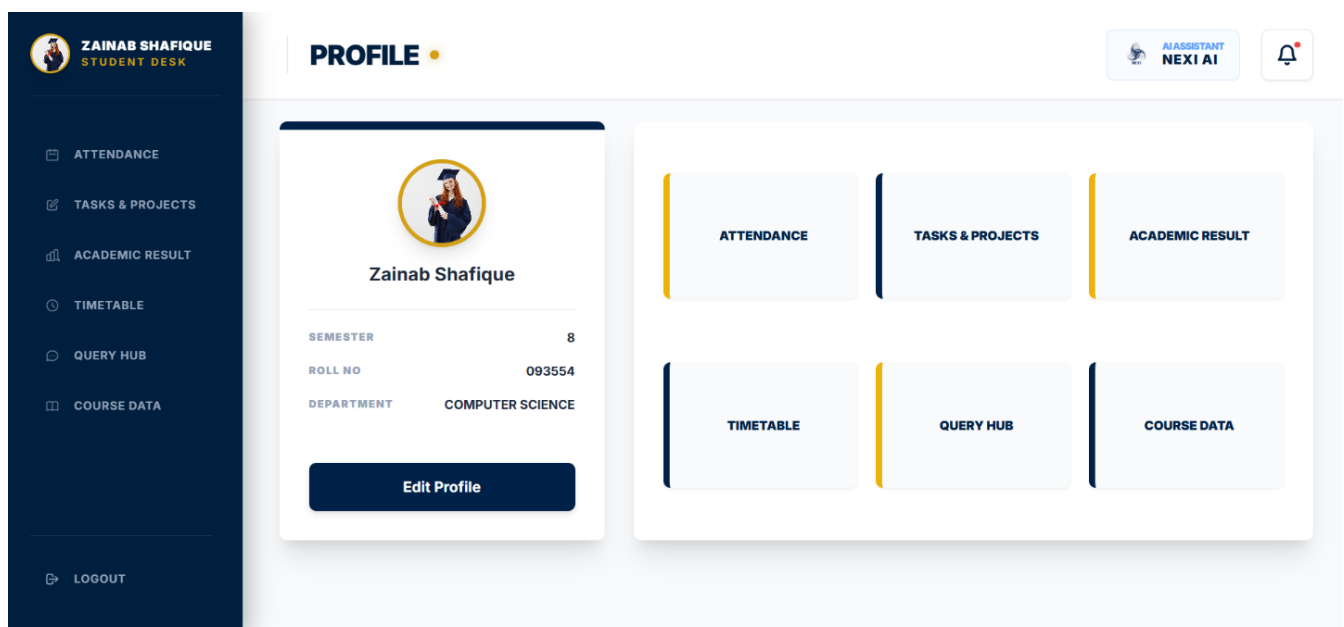
- Backend handles all student, course, attendance, and result data.
- Ensures data integrity and role-based access security.

6.2.11. Screenshots

1. Authentication Page



2. Student Dashboard



3.Attendance Page (Student)

ZAINAB SHAFIQUE
STUDENT DESK

ATTENDANCE

TASKS & PROJECTS

ACADEMIC RESULT

TIMETABLE

QUERY HUB

COURSE DATA

LOGOUT

←

ATTENDANCE

Summary

PRESENT
3

ABSENT
0

LATE
0

LEAVE
0

All Time

Download Log

DATE	STATUS
2026-07-02	PRESENT
2026-07-01	PRESENT
2026-06-12	PRESENT

Leave Management

+ Request Leave

Status & History

Urgent piece of work

REJECTED

Download

Edit

Delete

Reply: Processed by Class Incharge

Sick Leave

APPROVED

Download

Edit

Delete

Reply: Processed by Class Incharge

4.Course Data (Student)

ZAINAB SHAFIQUE
STUDENT DESK

ATTENDANCE

TASKS & PROJECTS

ACADEMIC RESULT

TIMETABLE

QUERY HUB

COURSE DATA

LOGOUT

←

DATA

ACADEMIC STATUS

2 COURSES FOUND

Live Data

Folder Icon

INFORMATION SECURITY

CODE: EC-441 • 3 FOLDERS

VIEW MATERIAL →

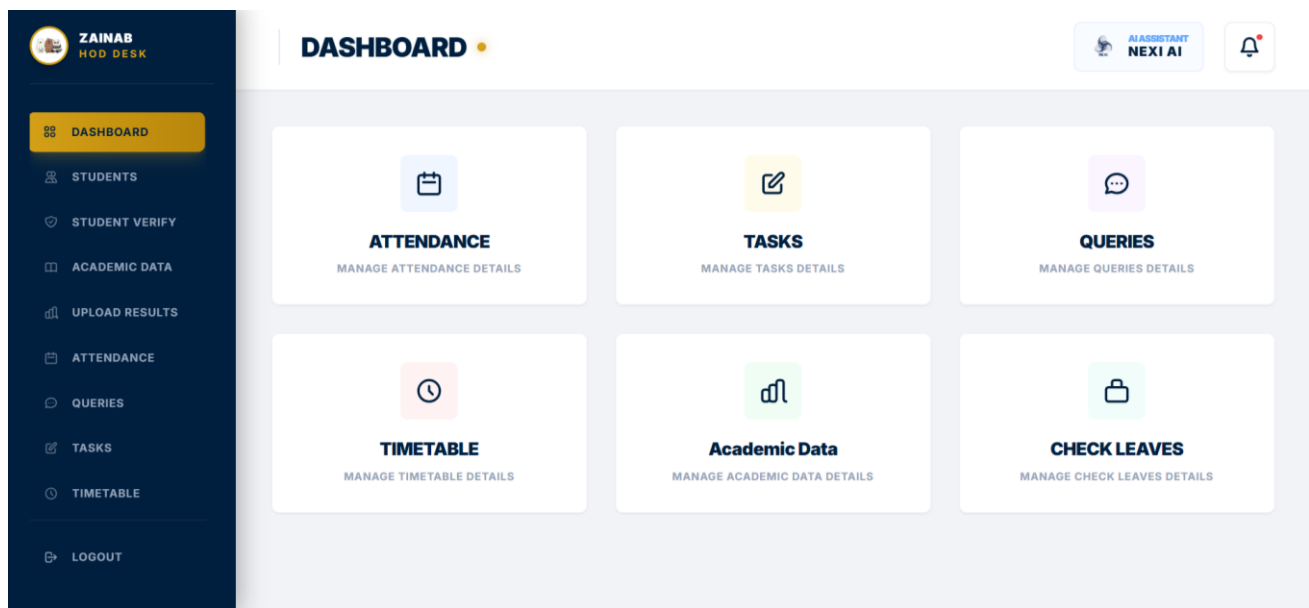
Folder Icon

THE HOLY QURAN

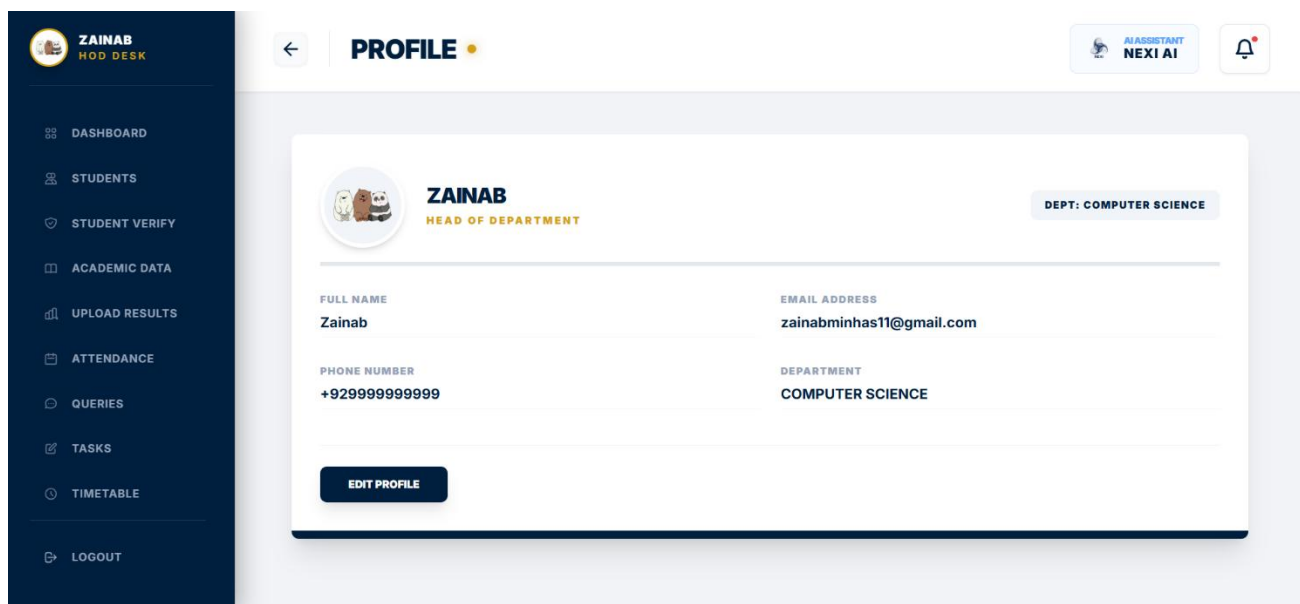
CODE: HQ-008 • 3 FOLDERS

VIEW MATERIAL →

5. Teacher Dashboard



6. Teacher Profile



7.Upload Result (For Students)

- DASHBOARD
- STUDENTS
- STUDENT VERIFY
- ACADEMIC DATA
- UPLOAD RESULTS**
- ATTENDANCE
- QUERIES
- TASKS
- TIMETABLE
- LOGOUT

UPLOAD RESULTS

Export Semester PDF

Save All Records

UNIVERSITY AFFILIATION

University of the Punjab

DEPARTMENT

COMPUTER SCIENCE

SEMESTER

1 Semester

LIVE STUDENT RECORD LEDGER (COMPUTER SCIENCE - 1)

ROLL #	STUDENT NAME	GPA	CGPA	DMC TRANSCRIPT	ACTIONS
333333	ZAI	4	4	Choose file No file chosen	
888888	ZAMI	3	3	Choose file No file chosen	

8.Mark Attendance

- DASHBOARD
- STUDENTS
- STUDENT VERIFY
- ACADEMIC DATA
- UPLOAD RESULTS
- ATTENDANCE**
- QUERIES
- TASKS
- TIMETABLE
- LOGOUT

ATTENDANCE

MAIN MENU

DEPARTMENT

COMPUTER SCIENCE

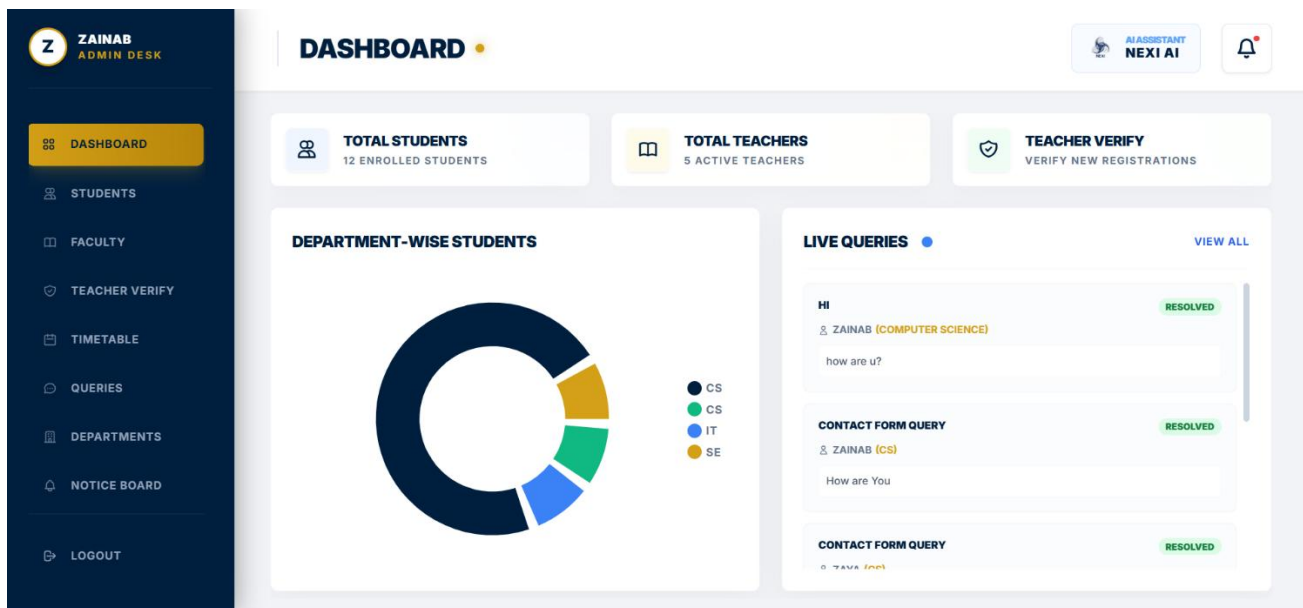
SEMESTER

1 Semester

ROLL NO	STUDENT NAME	STATUS ACTION
333333	ZAI	P A L
888888	ZAMI	P A L

SUBMIT ATTENDANCE SHEET

9.Admin Dashboard



10.Student Management

The Student Management page includes a sidebar with navigation options: DASHBOARD, STUDENTS, FACULTY, TEACHER VERIFY, TIMETABLE, QUERIES, DEPARTMENTS, NOTICE BOARD, and LOGOUT. The main content area features a 'STUDENT MANAGEMENT' header, a user profile for 'ZAINAB NEXI AI', and a search bar with '093'. Below the search bar is a table listing students with columns for ROLL NO, FULL NAME, DEPARTMENT, SEMESTER, and ACTIONS.

ROLL NO	FULL NAME	DEPARTMENT	SEMESTER	ACTIONS
093554	Zainab Shafique	COMPUTER SCIENCE	8	EDIT DELETE
093635	Bisma amjad	COMPUTER SCIENCE	8	EDIT DELETE
093631	sheri	COMPUTER SCIENCE	8	EDIT DELETE
093549	SamiUllah	COMPUTER SCIENCE	8	EDIT DELETE

11.Faculty Management

The screenshot shows the 'FACULTY MANAGEMENT' page. On the left is a dark blue sidebar with a 'ZAINAB ADMIN DESK' header and a menu including DASHBOARD, STUDENTS, FACULTY (highlighted), TEACHER VERIFY, TIMETABLE, QUERIES, DEPARTMENTS, NOTICE BOARD, and LOGOUT. The main content area has a top bar with a back arrow, 'FACULTY MANAGEMENT', and an AI Assistant 'NEXI AI' button. Below this, there are three faculty cards. Each card displays a profile picture, name, department, email, and phone number. The first card for 'Khadija' (COMPUTER SCIENCE) has a 'MAKE HOD' button. The second card for 'Zainab' (COMPUTER SCIENCE) has a 'HOD' badge and a 'REVOKE HOD' button. The third card for 'DMS' (BS ENGLISH) has a 'MAKE HOD' button. Each card also has edit and delete icons.

12.Handle Queries

The screenshot shows the 'QUERIES' page. The sidebar is identical to the previous screen. The main content area has a top bar with a back arrow, 'QUERIES', and an AI Assistant 'NEXI AI' button. Below the top bar are two filters: 'COMPUTER SCIENCE' and 'Student'. The queries are listed below. The first query is a 'CONTACT FORM QUERY' from 'Zainab (zainabminhas294@gmail.com)' in 'CS' department, 'SEM 1'. The query text is '"How are You"'. It has an 'EDIT REPLY' button and a green 'RESPONSE' box containing '"Fine Alhumdulillah"'. The status is 'RESOLVED'. The second query is a 'CONTACT FORM QUERY' from 'Zaya (zaya@gmail.com)' in 'CS' department, 'SEM 8'. It also has a 'RESOLVED' status.

13. Manage Departments

ZAINAB ADMIN DESK

DEPARTMENT MANAGEMENT

+ ADD DEPARTMENT

BS ARTIFICIAL INTELLIGENCE

ARTIFICIAL INTELLIGENCE (AI) CAN AUTOMATICALLY PERFORM REPETITIVE TASKS, HELPING SAVE TIME AND REDUCE HUMAN...

HIGHLIGHTS

- AI AUTOMATES REPETITIVE TASKS, SAVING TIME AND INCREASING EFFICIENCY.
- AI ANALYZES DATA TO PROVIDE ACCURATE INSIGHTS AND SUPPORT BETTER DECISIONS.

BS ECONOMICS

PROVIDES A DEEP UNDERSTANDING OF HOW MARKETS, ECONOMIES, AND FINANCIAL SYSTEMS WORK TO STUDY ECONOMIC TREND...

HIGHLIGHTS

- STRONG FOUNDATION IN MICROECONOMICS AND MACROECONOMICS
- DEVELOPMENT OF ANALYTICAL STATISTICS SKILLS

BS POL-SCIENCE

PROVIDES INSIGHT INTO POLITICAL SYSTEMS, GOVERNANCE, PUBLIC POLICIES, AND INTERNATIONAL RELATIONS FOR CRITICAL...

HIGHLIGHTS

- STUDY OF GOVERNMENT STRUCTURES AND GLOBAL AFFAIRS
- DEVELOPMENT OF LEADERSHIP AND DECISION-MAKING SKILLS

14. Notice Management

ZAINAB ADMIN DESK

NOTICE BOARD

CREATE NEW ANNOUNCEMENT

Notice Title...

Write full announcement here...

TARGET AUDIENCE:

☒ All ☐ Students ☐ Teachers

POST NOTICE

RECENT HISTORY

TO: STUDENT 03/07/2026

QUERY REPLIED: DBMS

Your message has received a reply.
Reply: primary key is a key which identify a row or a cou...

TO: TEACHER 03/07/2026

NEW LEAVE APPLICATION

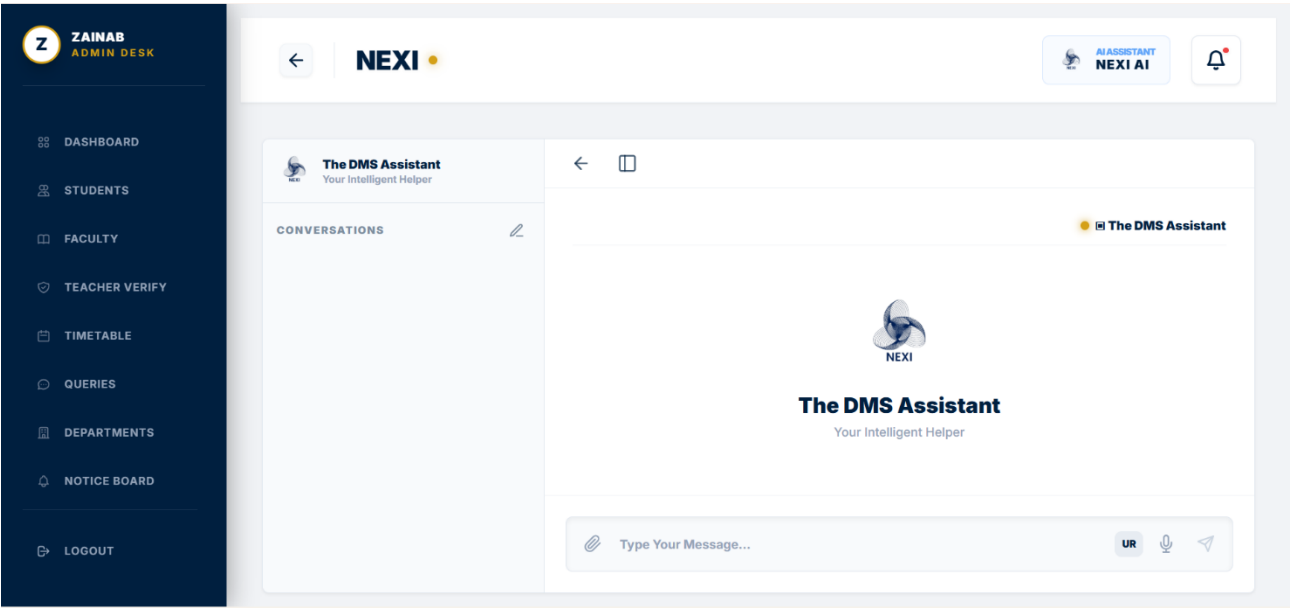
Bisma amjad has submitted a leave application: leave

TO: STUDENT 03/07/2026

LEAVE APPLICATION REJECTED

Your leave application for Urgent piece of work has been rejected. Remarks:

15.NEXI AI Chatbot



Chapter 7

Testing

7. TESTING

7.1 Testing Techniques Used

To ensure the Department Management System (DMS) meets all functional, usability, and performance standards, the following software testing techniques were applied:

7.1.1 Functional Testing

- **Purpose:** Ensures each feature of the system (Attendance, Grading, Enrollment) performs according to specifications.
- **Testing Focus:**
 - **Students:** Ability to view profiles, track attendance, and check grades.
 - **Faculty:** Ability to manage course materials, input attendance, and record grades.
 - **Admin:** Ability to manage users, courses, and department-wide reporting.

7.1.2 Usability Testing

- **Purpose:** Ensures the DMS is intuitive and easy to navigate for non-technical users.
- **Testing Focus:** Clear labels on forms, responsive layout for different devices, and logical flow between dashboard modules.

7.1.3 Performance Testing

- **Purpose:** Evaluates system efficiency under peak academic loads (e.g., result declaration days).

7.2 Test Cases (Samples)

7.2.1. Test Cases For Student Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
STU_TC_01	Registration	Form mein valid details bharna	Account successfully create hona	Pass
STU_TC_02	Login (2FA)	ID/Password + OTP enter karna	Dashboard load hona	Pass
STU_TC_03	View Attendance	Attendance tab par click karna	Sahi attendance % dikhna	Pass
STU_TC_04	View Results	Results tab par click karna	Semester-wise grades dikhna	Pass
STU_TC_05	Download Notes	Course material select karna	File ka download hona	Pass
STU_TC_06	Submit Query	Support form submit karna	Confirmation message milna	Pass

7.2.2. Test Cases For Faculty Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
FAC_TC_01	Admin Approval	Register and wait for Admin	Login access granted only after approval	Pass
FAC_TC_02	2FA Login	Enter credentials + OTP	Dashboard loads	Pass
FAC_TC_03	Upload Material	Select course and upload file	File saved in DB and viewable to students	Pass
FAC_TC_04	Mark Attendance	Update student status (P/A)	Attendance database syncs in real-time	Pass
FAC_TC_05	Grade Entry	Enter marks for subject	Marks saved & student record updated	Pass
FAC_TC_06	AI Assistance	Request assignment feedback	AI provides analytical suggestions	Pass

7.2.3. Test Cases For Admin Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
ADM_TC_01	Admin Login	Secure credentials + OTP	Admin dashboard loads successfully	Pass
ADM_TC_02	Approve Faculty	Select pending user -> 'Approve'	User account activated in Database	Pass
ADM_TC_03	Manage Course	Add new subject + credit hours	Course added to global list	Pass
ADM_TC_04	AI Analytics	Click on 'View Trends'	AI-generated performance reports show	Pass
ADM_TC_05	Resolve Issue	Select ticket -> 'Resolve'	Ticket status updated to Closed	Pass
ADM_TC_06	User Block	Select user -> 'Deactivate'	User login access revoked	Pass

7.3 Detailed Test Cases

7.3.1. Usability Testing (UT)

Test Case ID	Module	Test Scenario	Expected Result
UT_TC_01	Student	Find Attendance/Result	Easily accessible from dashboard
UT_TC_02	Student	View Course Material	Intuitive file download interface
UT_TC_03	Faculty	Grade Entry	Simple and clear input form
UT_TC_04	Faculty	Attendance Management	Calendar view is clear and clickable
UT_TC_05	Admin	User Management	Smooth navigation to user list
UT_TC_06	Admin	Report Generation	Clear filter and download options
UT_TC_07	All	Mobile Responsive	Interface adapts to mobile screens

7.3.2. Functional Testing (FT)

Test Case ID	Module	Test Scenario	Expected Result
FT_TC_01	Student	Registration & 2FA	Account created & OTP verified
FT_TC_02	Student	Login	Successful redirect to student dashboard
FT_TC_03	Student	View Grades/Attendance	Correct data fetched from database
FT_TC_04	Faculty	Upload Resource	File successfully saved in system
FT_TC_05	Faculty	Grade Entry	Marks updated in student records
FT_TC_06	Faculty	Attendance Entry	Attendance status reflected in DB
FT_TC_07	Admin	Approve Registration	Faculty account status: 'Active'
FT_TC_08	Admin	Course Assignment	Faculty linked to specific course
FT_TC_09	Admin	AI-Analytics	System displays performance trends
FT_TC_10	Admin	Resolve Query	Support ticket status: 'Closed'

7.3.3. Performance Testing (PT)

Test Case ID	Module	Test Scenario	Expected Result
PT_TC_01	Student	50+ Concurrent Login	Page loads in ≤ 2 seconds
PT_TC_02	Student	Result Access	Data fetched in ≤ 3 seconds
PT_TC_03	Faculty	Uploading Large Files	Upload completes in ≤ 5 seconds
PT_TC_04	System	Simultaneous Attendance Entry	No database bottleneck or lag
PT_TC_05	Admin	Report Generation	Monthly report ready in ≤ 10 seconds
PT_TC_06	AI	Data Analytics Processing	AI trend report ready in ≤ 12 seconds

7.3.4. Testing Results Summary

Test Case ID	Module	Scenario	Expected Result	Actual Result	Status
FT_TC_01	Auth	Login + 2FA	Access Granted	Success	Pass
FT_TC_02	Student	View Result	Data Displayed	Success	Pass
FT_TC_03	Faculty	Save Attendance	DB Updated	Success	Pass
FT_TC_04	Admin	Approve Faculty	Account Active	Success	Pass
FT_TC_05	Admin	AI Trend View	Chart Rendered	Success	Pass
FT_TC_06	System	Responsiveness	Adaptive UI	Success	Pass
FT_TC_07	Student	Submit Query	Ticket Created	Success	Pass
FT_TC_08	Faculty	Upload Notes	File Stored	Success	Pass
FT_TC_09	Admin	Bulk User Update	Records Updated	Success	Pass
FT_TC_10	System	Performance	Fast Load	Success	Pass

Chapter 8

System Conversion

8. System Conversion

8.1. Conversion Method

Chosen Method: Direct Conversion

We will transition from the existing manual record-keeping process to the **Department Management System (DMS)** through a direct conversion strategy. This approach is ideal for academic evaluation, as it allows the entire system to become operational simultaneously, ensuring a seamless data transition without operational interruptions.

8.2. Training: User Guide for Core Modules

8.2.1. Student Panel

- **Access:** Secure sign-up and login facilitated by 2FA (OTP) verification.
- **Academic Monitoring:** A real-time dashboard view of attendance percentages and semester-wise results.
- **Resources:** Access to a digital library for downloading course materials and PDFs provided by faculty.
- **Support:** An integrated ticketing system to submit academic queries or issues to the Admin or Faculty.

8.2.2. Faculty Panel

- **Account Activation:** Following registration, faculty members wait for administrative approval to ensure system security.
- **Academic Duties:** Tools to manage daily attendance and input student grades.
- **Content Management:** A streamlined process to upload course materials for student access.
- **AI Support:** Utilization of an AI-based feedback system to analyze assignments and provide grading suggestions.

8.2.3. Admin Panel

- **User Oversight:** Authority to review, verify, or approve registration requests for students and faculty.
- **Monitoring:** Access to AI-driven analytics dashboards to track department-wide performance trends.
- **System Admin:** Capability to resolve support tickets and manage overall department settings.

8.3. Challenges and Solutions

Challenge	Solution
Data Synchronization	Implemented a centralized database to ensure real-time data consistency.
User Onboarding	Created a user-friendly guide with clear steps for 2FA-based secure login.
Role-based Security	Enforced strict access controls to ensure users only access their relevant modules.
System Scalability	Developed a modular architecture to allow easy updates for new courses and students.
AI Accuracy	Conducted continuous testing of AI models against academic data to ensure precise feedback.

Chapter 9

Conclusion

9. CONCLUSION

9.1. Evaluation

Objective	Status
Implement Student, Faculty, and Admin authentication with 2FA/OTP	✓ Completed
Enable students to view real-time attendance and semester results	✓ Completed
Implement secure file upload for course materials	✓ Completed
Allow faculty to manage course listings, grades, and attendance	✓ Completed
Admin capability to approve/reject user registrations	✓ Completed
Admin dashboard with AI-based performance analytics	✓ Completed
Integration of AI to provide study planning and feedback suggestions	✓ Completed
Secure backend with JWT authentication and role-based access	✓ Completed
Deployment of all panels on a cloud environment	✓ Completed

9.2. Traceability Matrix

Req ID	Requirement Description	Design Component	Code Modules	Test ID
DMS-01	Student/Faculty/Admin Auth (2FA)	AuthController, OTPManager	auth.controller.js, otp.js	FT_TC_01
DMS-02	View Attendance & Results	StudentDashboard	student.controller.js, Results.jsx	FT_TC_02
DMS-03	Faculty Resource Management	CourseController	course.controller.js, FileUpload.jsx	FT_TC_04
DMS-04	AI-based Analytics	AdminDashboard	ai.service.js, analytics.controller.js	ADM_TC_04
DMS-05	Admin User Approval	AdminPanel	user.controller.js, Approval.jsx	ADM_TC_02
DMS-06	Real-time Attendance Entry	AttendanceModule	attendance.controller.js	FAC_TC_03
DMS-07	Support Ticketing System	SupportModule	ticket.controller.js, Support.jsx	STU_TC_06

Summary

The **Department Management System (DMS)** is a comprehensive, full-stack web application designed to digitize academic operations. The platform facilitates seamless interaction between students, faculty, and administration, covering essential academic needs such as attendance tracking, result management, secure resource sharing, and grievance redressal.

The system features a modular architecture developed using **React.js** for the frontend and **Node.js/Express.js** for the backend, with **MongoDB Atlas** ensuring scalable data storage. Security is a top priority, enforced via **JWT authentication** and **2FA (OTP verification)**. Furthermore, the integration of **AI-driven analytics** provides actionable insights for administration to monitor academic trends, making the DMS a smart, secure, and scalable solution for modern educational institutions.

References

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